Gene Expression Variability in Human Induced Pluripotent Stem Cells

BIOL 806 Final Project

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Abstract

Human induced pluripotent stem cells (iPSCs) are widely used to model human development, disease, and for potential cell therapies. Accurate characterization of iPSC lines is essential to ensure experimental and manufacturing consistency, including assessing pluripotency, differentiation, and reference gene expression. However, previous studies have shown that commonly used housekeeping genes, such as GAPDH, can vary across iPSC lines, complicating normalization.

Here, we analyzed publicly available RNA-seq data from 317 iPSC samples from 101 diverse donors to quantify RNA composition, identify drivers of gene expression variability, and evaluate pluripotency, differentiation, and housekeeping gene expression. While multiple RNA types were detected, 91.7% of total expression originated from protein-coding mRNAs. Principal component analysis (PCA) and Welch's one-way ANOVA revealed that donor age–sex groups and race contributed significantly to expression variability, with sex-linked genes among the top drivers. Pluripotency markers, such as LIN28A and SOX2, and differentiation markers, such as SOX17 and CD34, amongst many other genes, exhibited variable expression across donor groups. Notably, housekeeping genes (GAPDH, HPRT1, PGK1) also significantly varied, highlighting limitations for their use as universal references.

These results highlight that donor age, sex, and race can significantly influence iPSC transcriptomes, affecting the expression of many genes that are commonly used as references and control markers. This displays the need to account for donor demographics when interpreting gene expression data, selecting reference genes, or designing experiments using iPSCs, to ensure accurate and reproducible results.

Introduction

The discovery of human induced pluripotent stem cells (iPSCs) by Takahashi and Yamanaka in 2007 (Takahashi et al., 2007) has revolutionized the regenerative medicine field. iPSCs are created by reprogramming somatic cells into an embryonic like state using the Yamanaka transciption factors: SOX2, OCT4, KLF4, and c-Myc. These cells have vast capabilities due to their ability to model human development and diseases *in vitro*, and their potential for patient specific (autologous) and "off the shelf" (allogeneic) cell therapies (Cerneckis et al., 2024).

With that being said, the characterization of iPSC lines is essential in any experimental or manufacturing process to ensure consistency in culturing. This includes analyzing the expression of pluripotency (stem cell) markers, differentiation markers, and potential chromosomal abnormalies, to access the population of the culture. This is commonly done using techniques such as reverse transcriptase or real time quantitative PCR (RT-qPCR) to analyze mRNA expression (Artyukhov et al., 2017), or using an antibody based experiment to analyze protein expression levels, such as Flow Cytometry or immunocytochemistry. When conducting RT-qPCR and similar experiments, reference genes, also known as housekeeping genes, are used to normalize the expression of other genes in a sample (Artyukhov et al., 2017). This allows the expression of genes to be compared across different samples, since it is presumed that the expression of the housekeeping gene is consistent across all of the samples. In stem cell samples, however, it has been found that the commonly used housekeeping genes, such as GAPDH, appear to fluctuate in iPSC samples (Panina et al., 2020), making it difficult to achieve accurate normalization data.

RNA-Seq is a technique that sequences the entirety of RNA molecules in a given sample, rather than using gene specific primers or probes as in RT-qPCR and other techniques. This allows the entire transciptome to be analyzed, providing a complete picture of the expression of genes in a sample (Wang et al., 2009). Additionally, gene expression is calculated by dividing the raw counts of a gene by the total counts in a sample, eliminating the need for a housekeeping gene. This allows for an unbiased and quantitiative analysis of millions of genes in sample (Wang et al., 2009). Despite these advantages, RNA-Seq is far more time consuming and expensive than RT-qPCR and other experiments, making it less practical in a manufacturing setting. Therefore, utilizing RNA-Seq to analyze a large number of iPSC samples could provide insight into housekeeping gene expression, allowing researchers to select ideal reference genes for their analysis.

A similar study, conducted by Carcamo-Orive *et al.,* investigated the variability of gene expression across 317 iPSC lines generated from 101 individuals (Carcamo-Orive et al., 2017) and created a publically avaible data set of their raw RNA Seq data. They found that genetic variability in the iPSC lines was influenced by a donor specific gene expression patterns, specifically influenced by donor age, sex, ancestry, and reprogramming variabilies. iPSCs can also be reprogrammed from a variety of somatic cells, such as blood, skin fibroblasts, hair keratinocytes, urine cells, mesencymal stem cells, and bone marrow (Poetsch et al., 2022).

While some somatic cells are easier to obtain than others, such as blood cells and fibroblasts, iPSCs from these lines are associated with higher genomic mutations due to the somatic cells' high turnover rate. Additionally, iPSCs appear to have an epigenetic memory of their tissue of origin, which has been found to influence their differentiation (Poetsch et al., 2022). All of these factors, and many others, could be contributing to the variations in gene expression in iPSC samples.

This publicly available dataset from Carcamo-Orive *et al.* was used to analyze the expression of RNA across iPSC samples, with a focus on the expression of pluripotency markers, differentiation markers, and housekeeping genes.

Methods

All scripts used for processing, filtering, visualization, and statistical analysis are available at: https://github.com/tfm1017/BIOL806-Final-Project.

Data Cleaning

RNA-Seq counts per million data (named: cpm_data), and sample metadata (named: raw_counts) were acquired from Stemformatics using the Carcamo-Orive (2017) dataset (Carcamo-Orive et al., 2017). Genes in this study were referred to by their Ensembl ID, so gene information, such as name and RNA type, was integrated from the Ensembl database. Using R (R Core Team, 2025), the raw count per million data was converted to a log base 2 scale with a 0.5 offset to prevent samples with a count of 0 from becoming skewed. Values for age, from the metadata file, were grouped by decade, and an additional column was added for an age + sex category (i.e., 40s female). Information from the sample description was then extracted to create columns for race and insulin resistance. The metadata and log counts per million files were then merged into one data frame for further analysis.

Category Tables

The number of samples for each group within the age-sex and race categories were summarized and displayed as two tables using the knitr (Xie, 2025) and kableExtra (H. Zhu, 2024) packages.

Pie Charts by RNA Category

Genes detected in the RNA Seq counts per million data were grouped into the following categories: Protein coding, long non-coding RNA (lncRNA), Small RNAs, Pseudogenes, and Other. A count of the number of genes in each category was then performed by using the dplyr package (Wickham et al., 2023) from tidyverse (Wickham, 2023). The total expression of each category was also calculated using the raw count per million data. The scales (Wickham, Pedersen, et al., 2025) package from ggplot2 (Wickham, Chang, et al., 2025) was used to create two pie charts, one of the gene counts and one of the gene expression by RNA category, plotted together using the patchwork package (Pedersen, 2025).

Principal Component Analysis (PCA) Plots

Log2 expression values were first cleaned by removing entries without gene information and duplicate gene names. Gene variances were calculated across the samples, and genes with no variance were removed. To visualize the major sources of variation in the expression data, a principal component analysis (PCA) was performed using the FactoMineR package (Husson et al., 2025), which identifies the major axes of variation in the dataset. To assess if sample characteristics (sex, age decade, combined age sex groups, and race) influenced the PCA's structure, a Welch's one-way ANOVA, which does not assume equal variance, was performed separately for PCA1 and PCA2. Significant differences were found for the age sex groups and race categories, so two PCA biplots were generated, each overlaying confidence ellipses corresponding to either age sex groups or race. On both of the biplots, the top twenty genes with the highest variance across all samples were labeled and displayed as vectors. Additionally, PCA data was plotted as four scatterplots, colored by age, sex, age-sex, and race.

Pluripotency, Differentiation, and Housekeeping Gene Expression

The log2 expression of genes, categorized as pluripotency markers, differentiation markers, and housekeeping genes, was compared across sample groups (age-sex and race) using heatmaps, linear mixed-effects models, and Welch's ANOVA with post hoc Games-Howell tests. Heatmaps of log2 gene expression were generated using the pheatmap package (Kolde, 2025) to visualize overall expression patterns within each gene category. Linear mixed-effect models were then fit using lme4 (Bates et al., 2025) and lmerTest (Kuznetsova et al., 2020) to evaluate differences in gene expression across age, sex, age-sex ,and race groups, while accounting for gene-specific baseline expression levels by using random intercepts for genes. Percent differences in overall gene expression were calculated by exponentiating the estimated log2 scale coefficient back to fold change, subtracting one, and multiplying by 100. QQ plots and histograms were generated, but not shown here, for each model to view residuals and assess model assumptions. To identify which genes displayed statistically significant differences in expression between sample categories, Welch's one-way ANOVA, assuming unequal variance, was performed for each gene. P values were adjusted using the Benjamini-Hochberg method due to the high number of tests that were performed. For genes with significant ANOVA results, Games-Howell post hoc tests, using rstatix (Kassambara, 2025), were performed and p-values were Benjamini-Hochberg corrected. Significant Games Howell comparisons (adjusted $p < 0.05$) were annotated on bar plots displaying log2 expression for each gene across sample groups.

Results

Table 1: Sample counts by Age-Sex

Age-Sex Group	Sample Count
40s female	34
40s male	8
50s female	63
50s male	41
60s female	77
60s male	57
70s female	17
70s male	11

Table 2: Sample counts by Race

Race	Sample Count
African American	20
Asian	33
Caucasian	212
Hispanic	14
South Asian	13
NA	16

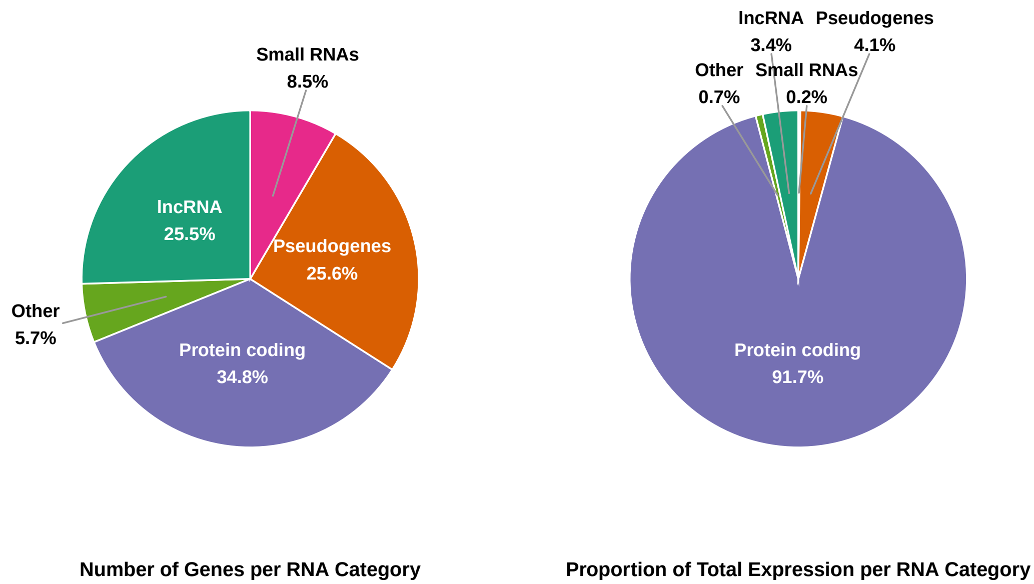


Figure 1: Detected RNA Types in iPSC Samples Analzyed using RNA Seq. The number of genes detected in each RNA category from RNA-seq data were counted and plotted as a pie chart (left). Small categories (< 2 %), such as rRNA, tRNA, and immunoglobulin genes, were combined into an 'Other' category. The expression of genes, using raw count per million data, was averaged by type of RNA and plotted as a pie chart (right).

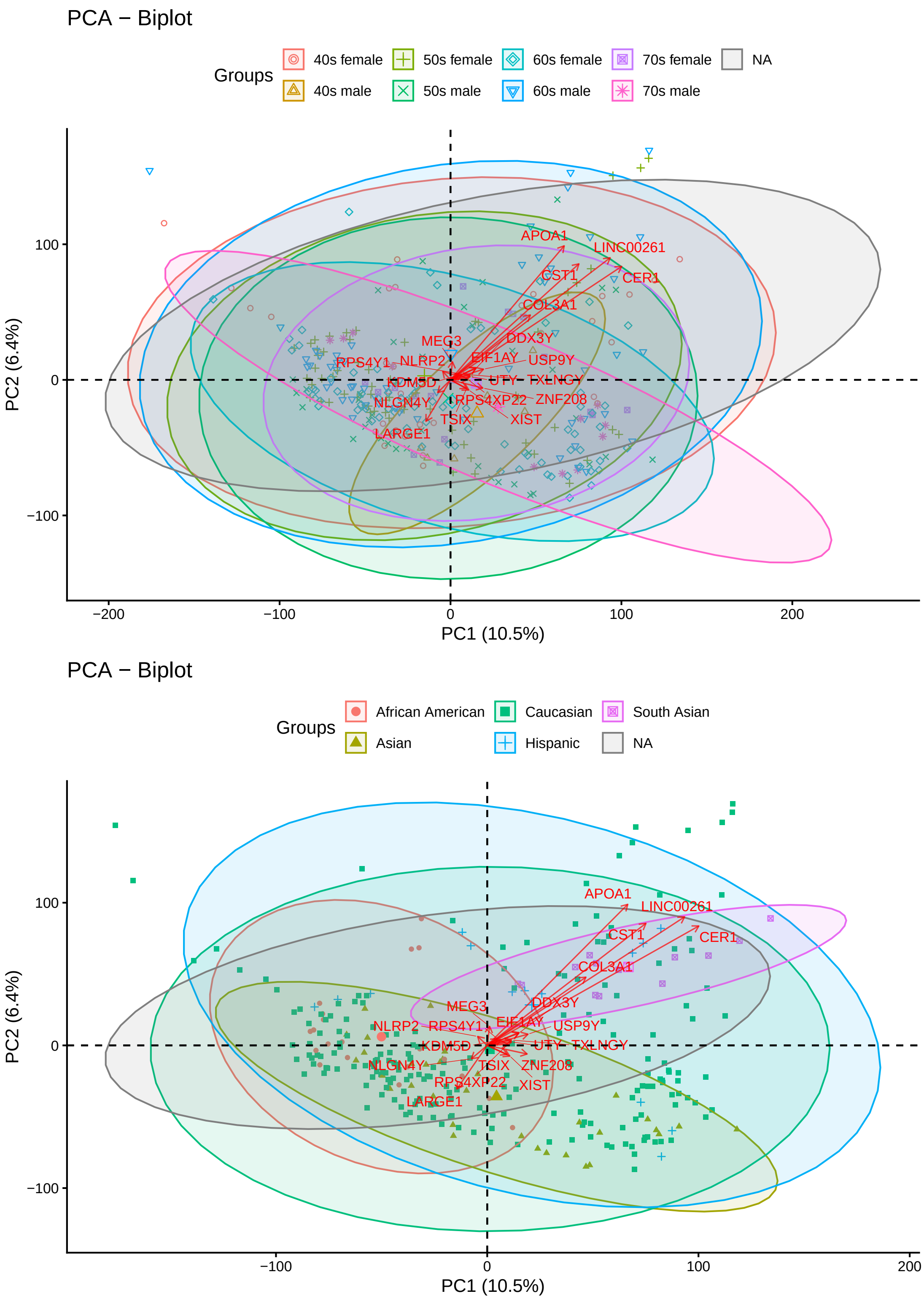


Figure 2: Principal Component Analysis (PCA) grouped by Donor's Age and Sex (top) and Race (bottom). PCA was performed using log2 gene expression across all samples, after removing genes with zero variance. PCA1 and PCA2 explain 10.5% and 6.4% of the variance in gene expression, respectively. Sample are colored by age-sex or race groups and ellipses represent 95% confidence intervals for each group. Two twenty genes contributing most to the variance, and thus most strongly influencing PCA1 and PCA2, are labeled as vectors.

Discussion

RNA Composition

The RNA Seq data is composed of a variety of different types of RNA, as seen in Figure 1. Of the genes detected, 34.8% of read genes are mRNA protein coding genes, whereas the remaining 65.2% are not protein coding and are primary composed of noncoding RNAs and pseudogenes, which are RNA sequences that look like protein coding genes but are not able to be translated into a protein. Specifically, 25.6% of the detected genes were pseuodegenes, 25.5% were long non coding RNA sequences and 8.5% were small RNAs, which included snoRNA, snRNA, scaRNA, sRNA, and ribozyme RNA. 5.7% were "Other" RNAs, which include ribosomal RNA, immunoglobin genes, T-cell receptor genes, and mitocondrial RNAs.

The right pie chart, on the other hand, displays the cumulative expression of these groups. For each gene in each sample, expression was reported as counts per million by Carcamo-Orive *et al.*. This is typically done by dividing the raw count of the gene by the total reads in the sample and multiplying by one million. The total expression was summed for each category, and this was graphed in Figure 2, which displays that 91.7% of the expression is from protein coding genes. The remaining 8.3% accounts for the expression of small RNAs, long non coding RNA, pseuodgenes, and "Other" genes. This is supported by Zhu *et al.*, who identfied that 93.7% of the transcripts in various RNA-Seq analysed tissue samples were protein coding, while 6.3% were non-coding transcripts (J. Zhu et al., 2016).

PCA

Principal component anaysis (PCA) was used to identify what genes expressed the greatest variation in expression of the iPSC analyzed samples and to visualize how demographic variables contribute to their variation. Figure 2 displays two PCA biplots annotated with 95% confidence ellipses for age-sex groups (top) and race groups (bottom). While the axis represent a low percentage of variability, 10.5% for PCA1 and 6.4% for PCA2, this is expected when a large number of genes (58,302), and thus a large number of variables, are included in the analysis. The ellipses displayed substantial overlap for the age-sex groups, but race based ellipses displayed some clustering of samples, suggesting that race contributes more strongly to the overall variations in gene expression of the samples, seen in Figure 2. The PCA biplots were then plotted as scatterplots and colored by age, sex, age-sex, and race to further view clustering in the dataset, as seen in Figure 3. This interpretation was quantified using a one way analysis of means, not assuming equal variance due to the variation in the number of samples per group, seen in Table 1 and 2. When tested independently, neither age nor sex was significantly associated with PC1 or PC2. However, when individuals were grouped by age–sex categories, a significant effect emerged (Welch's ANOVA, PCA1 $p = 0.199$ and PCA2 $p = 0.265$), suggesting that specific combinations of age and sex capture expression patterns not explained by either variable alone. Additionally, PCA1 and PCA2 were strongly influenced by

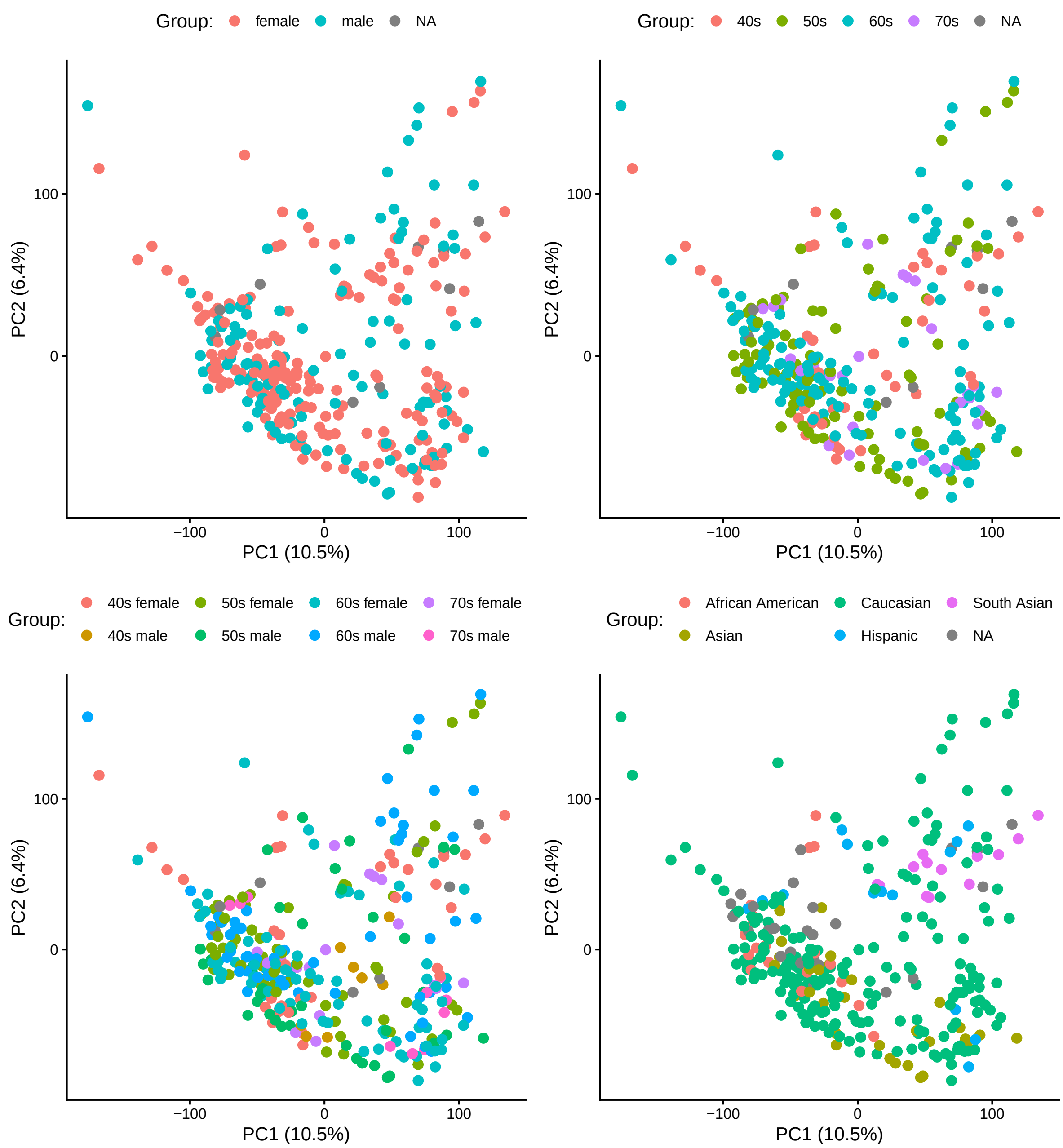
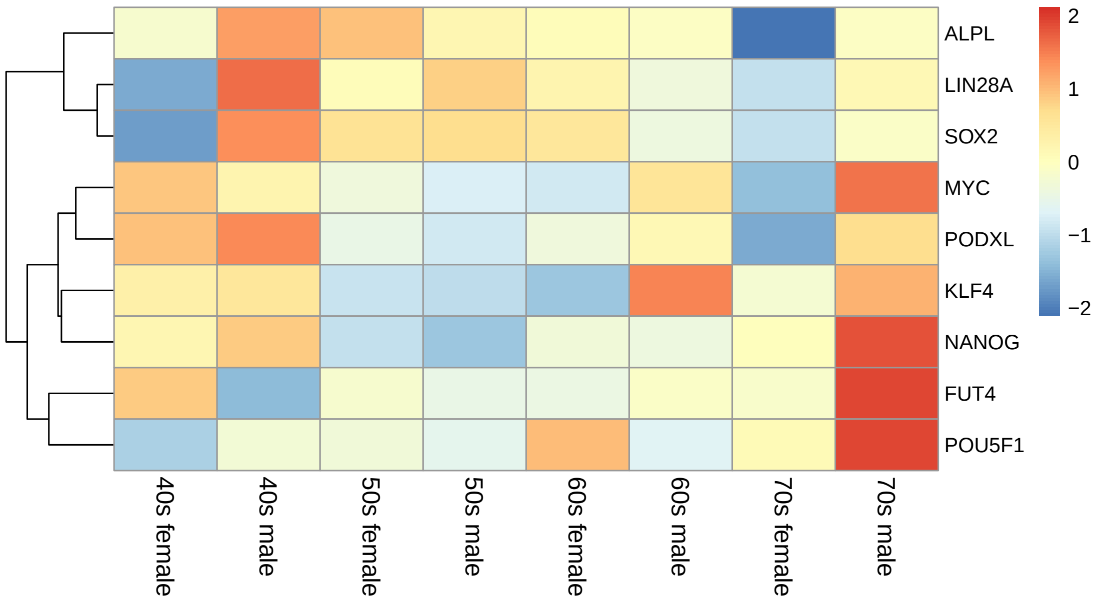


Figure 3: Principal Component Analysis (PCA) Grouped by Sex, Age, Age-Sex, and Race of Donors. PCA was performed using log2 gene expression across all samples, after remvoing genes with zero variance. PCA1 and PCA2 explain 10.5% and 6.4% of the variance in gene expression, respectively. The samples were then colored by catagory: sex, age, age-sex, and race, to observe clustering of gene expression.



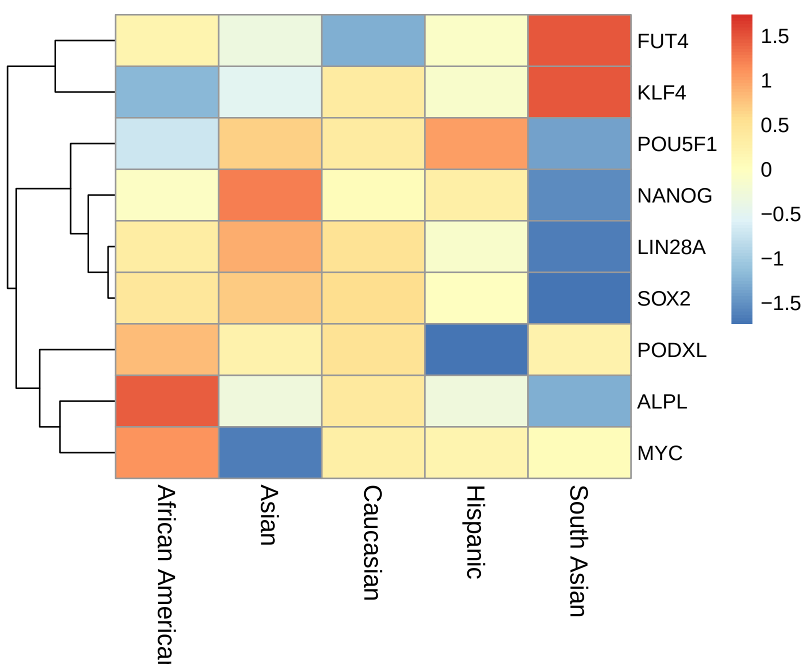


Figure 4: Heatmap of Pluripotency Gene Expression. Rows correspond to genes and columns correspond to samples grouped by age-sex (top) and race (bottom).Colors indicate relative expression levels based on log2 count per million expression. Hierarchical clustering was applied to the genes to highlight patterns of expression.

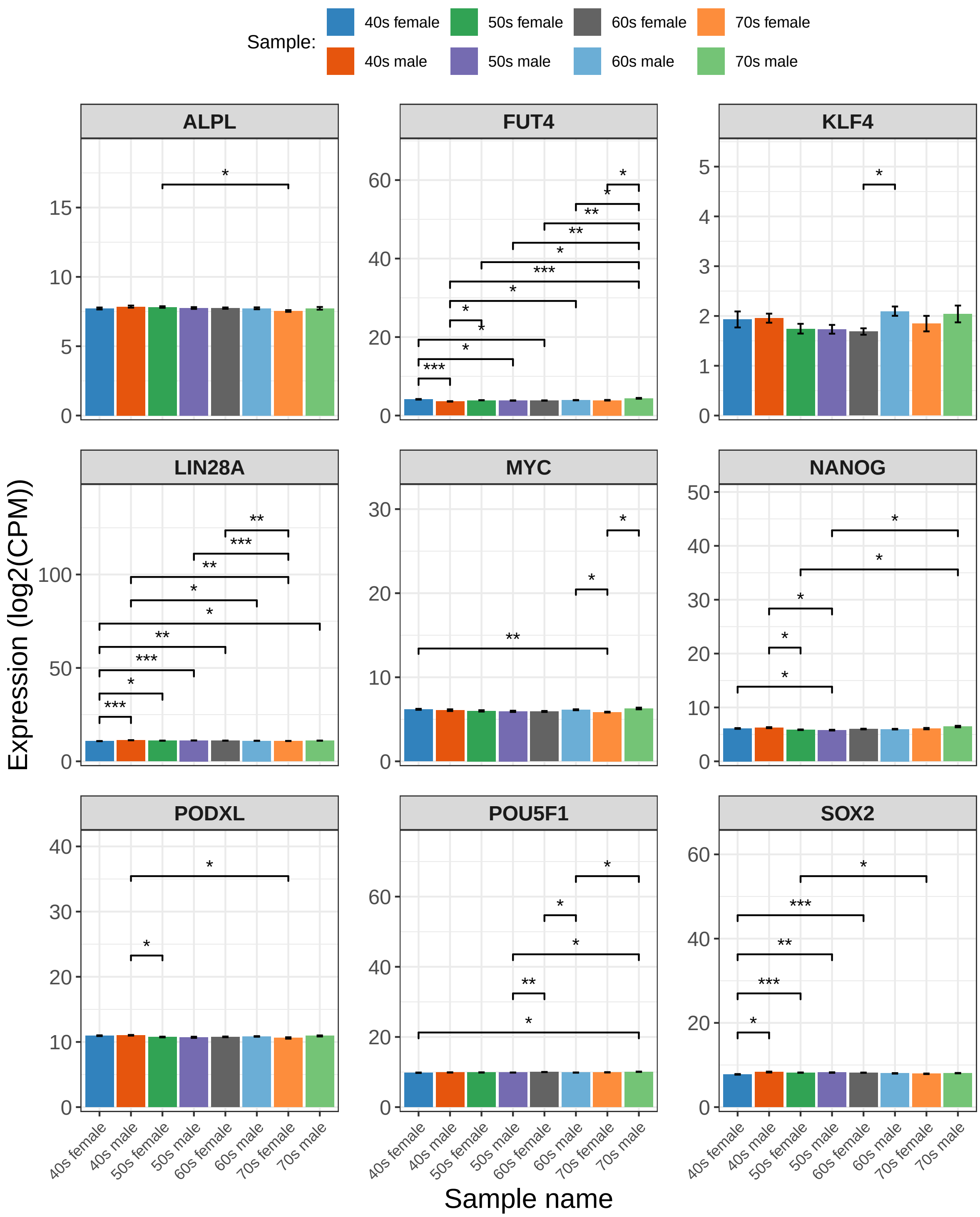


Figure 5: Expression of Pluripotency Genes by Age-Sex. Bar plots of log2 count per million gene expression by pluripotency gene. Bars represent mean expression ± standard error. Pairwise statistical significance was determined using Games-Howell post hoc tests following Welch's ANOVA, and significance was indicated with asterisks: $p < 0.05$ ($*$), $p < 0.01$ ($**$), $p < 0.001$ ($***$).

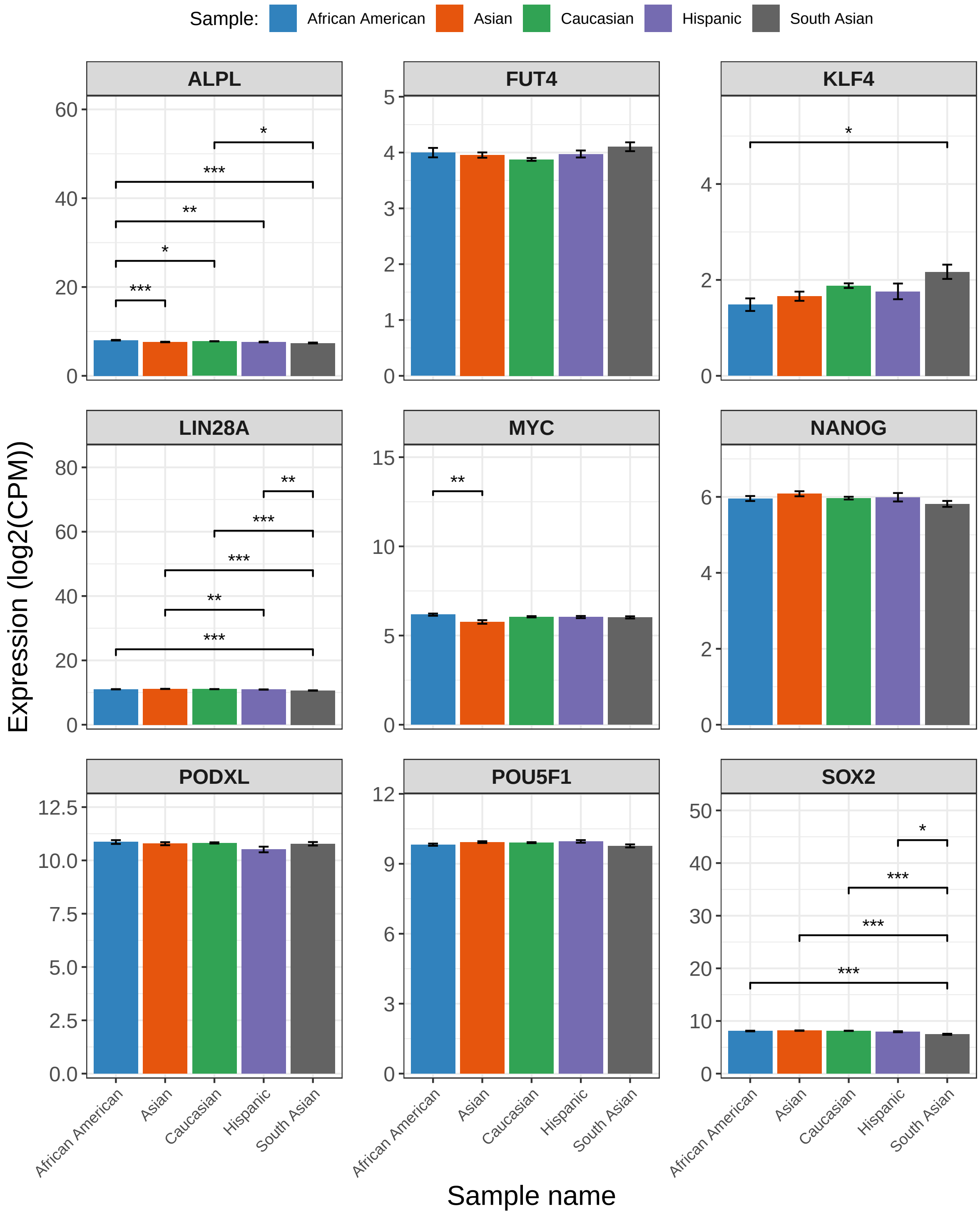
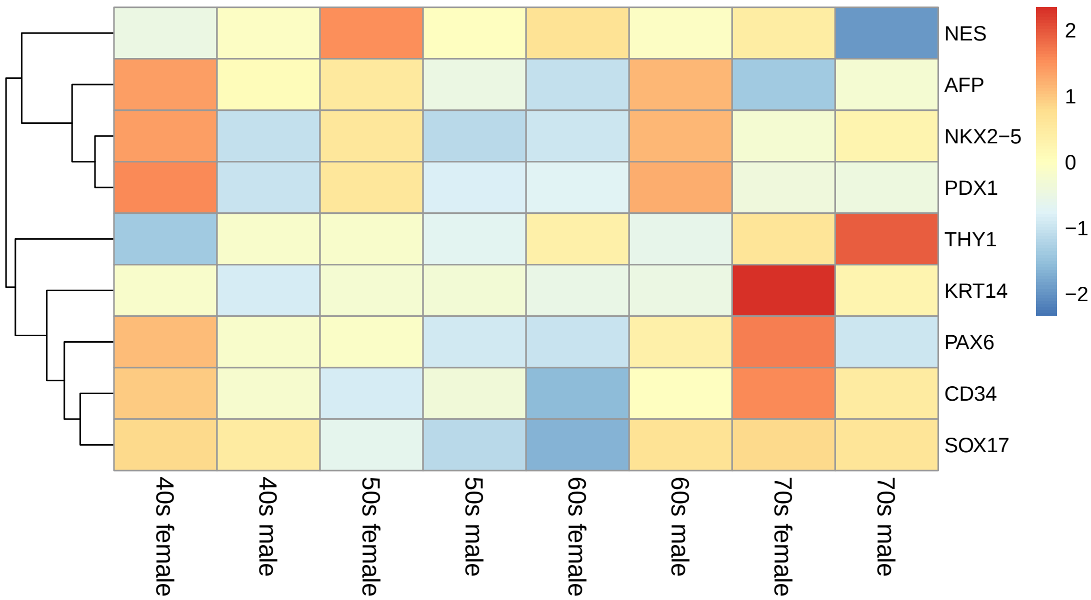


Figure 6: Expression of Pluripotency Genes by Race. Bar plots of log2 count per million gene expression by pluripotency gene. Bars represent mean expression ± standard error. Pairwise statistical significance was determined using Games-Howell post hoc tests following Welch's ANOVA, and significance was indicated with asterisks: $p < 0.05$ (∗), $p < 0.01$ (∗∗), $p < 0.001$ (∗∗∗)s



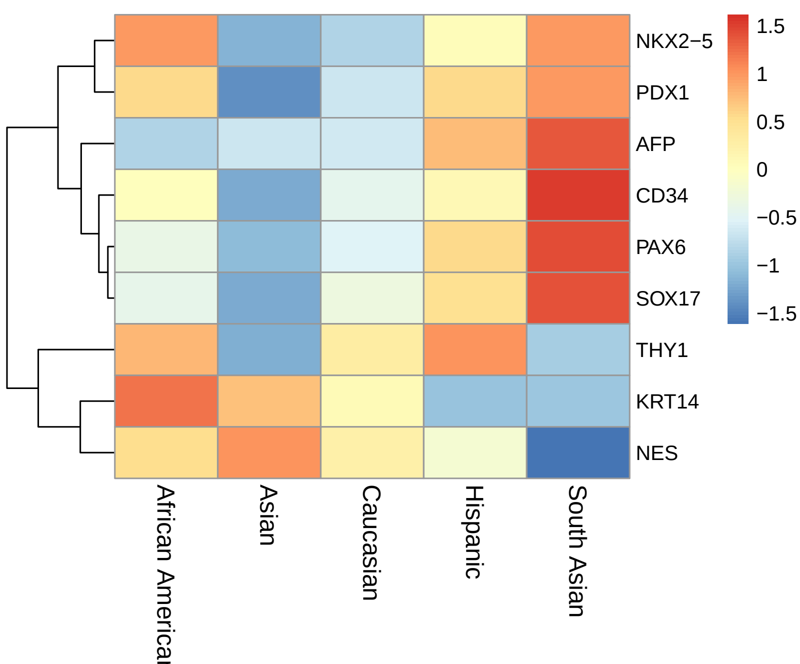


Figure 7: Heatmap of Differentiation Gene Expression. Rows correspond to genes and columns correspond to samples grouped by age-sex (top) and race (bottom).Colors indicate relative expression levels based on log2 count per million expression. Hierarchical clustering was applied to the genes to highlight patterns of expression.

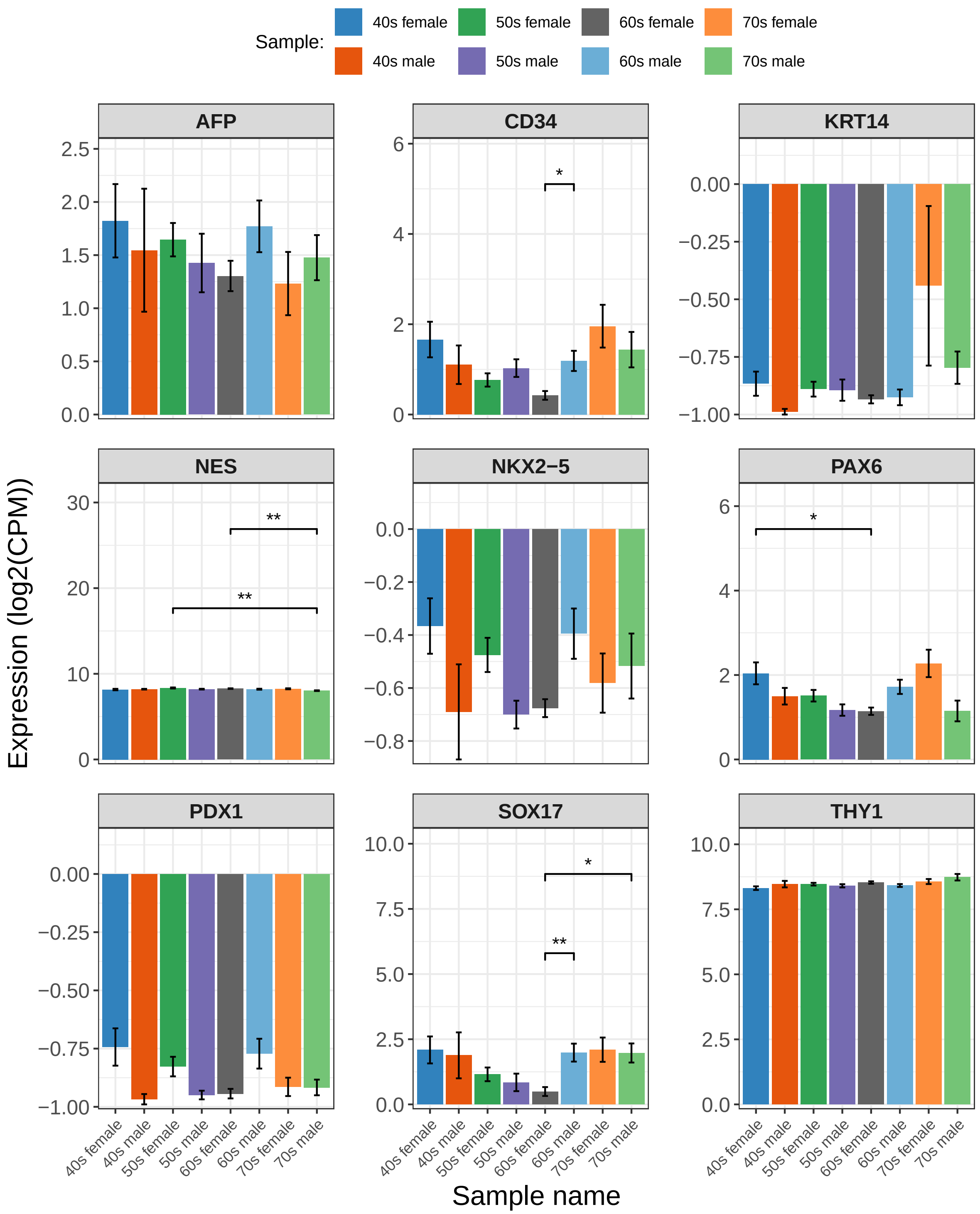


Figure 8: Expression of Differentiation Genes by Age-Sex. Bar plots display log2 count per million gene expression by differentiation gene. Bars represent mean expression $\pm$ standard error. Pairwise statistical significance was determined using Games-Howell post hoc tests following Welch's ANOVA, and significance was indicated with asterisks: $p < 0.05$ (∗), $p < 0.01$ (∗∗), $p < 0.001$ (∗∗∗).

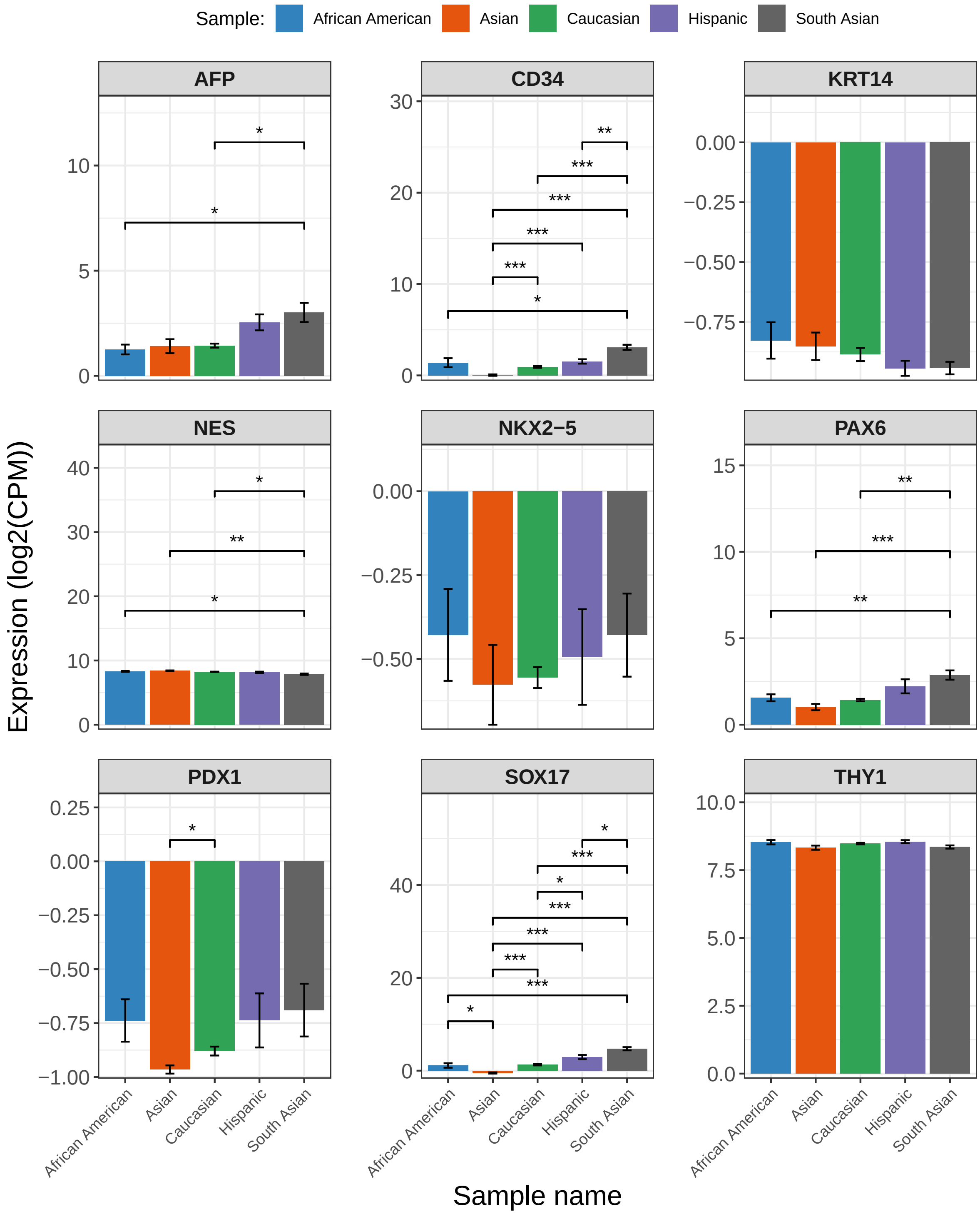


Figure 9: Expression of Differentiation Genes by Race. Bar plots display log2 count per million gene expression by differentiation gene. Bars represent mean expression ± standard error. Pairwise statistical significance was determined using Games-Howell post hoc tests following Welch's ANOVA, and significance was indicated with asterisks: $p < 0.05$ (∗), $p < 0.01$ (∗∗), $p < 0.001$ (∗∗∗).

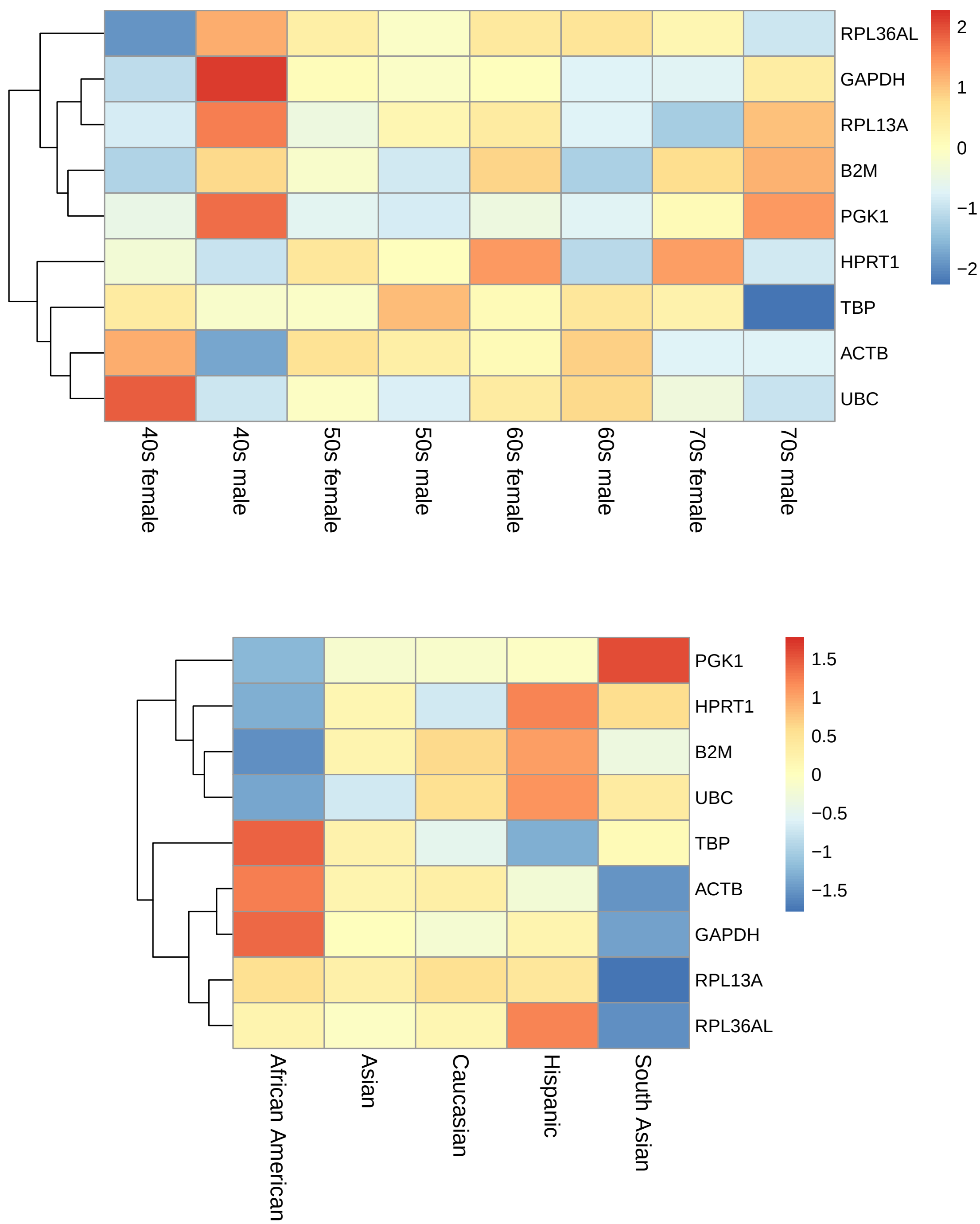


Figure 10: Heatmap of Housekeeping Gene Expression. Rows correspond to genes and columns correspond to samples grouped by age-sex (top) and race (bottom).Colors indicate relative expression levels based on log2 count per million expression. Hierarchical clustering was applied to the genes to highlight patterns of expression.

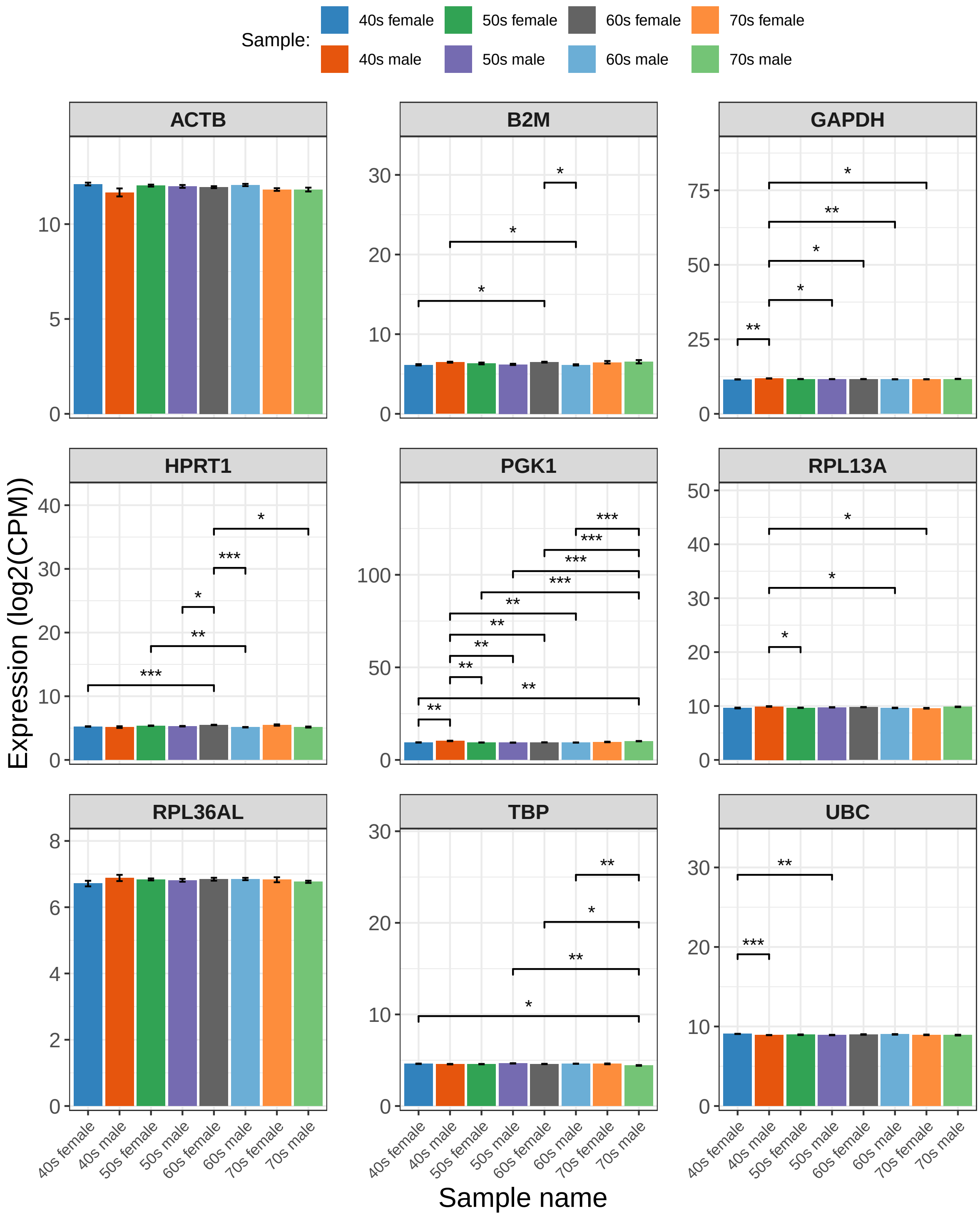


Figure 11: Expression of Housekeeping Genes by Age-Sex. Bar plots display log2 count per million gene expression by housekeeping gene. Bars represent mean expression $\pm$ standard error. Pairwise statistical significance was determined using Games-Howell post hoc tests following Welch's ANOVA, and significance was indicated with asterisks: $p < 0.05$ ($*$), $p < 0.01$ ($**$), $p < 0.001$ ($***$).

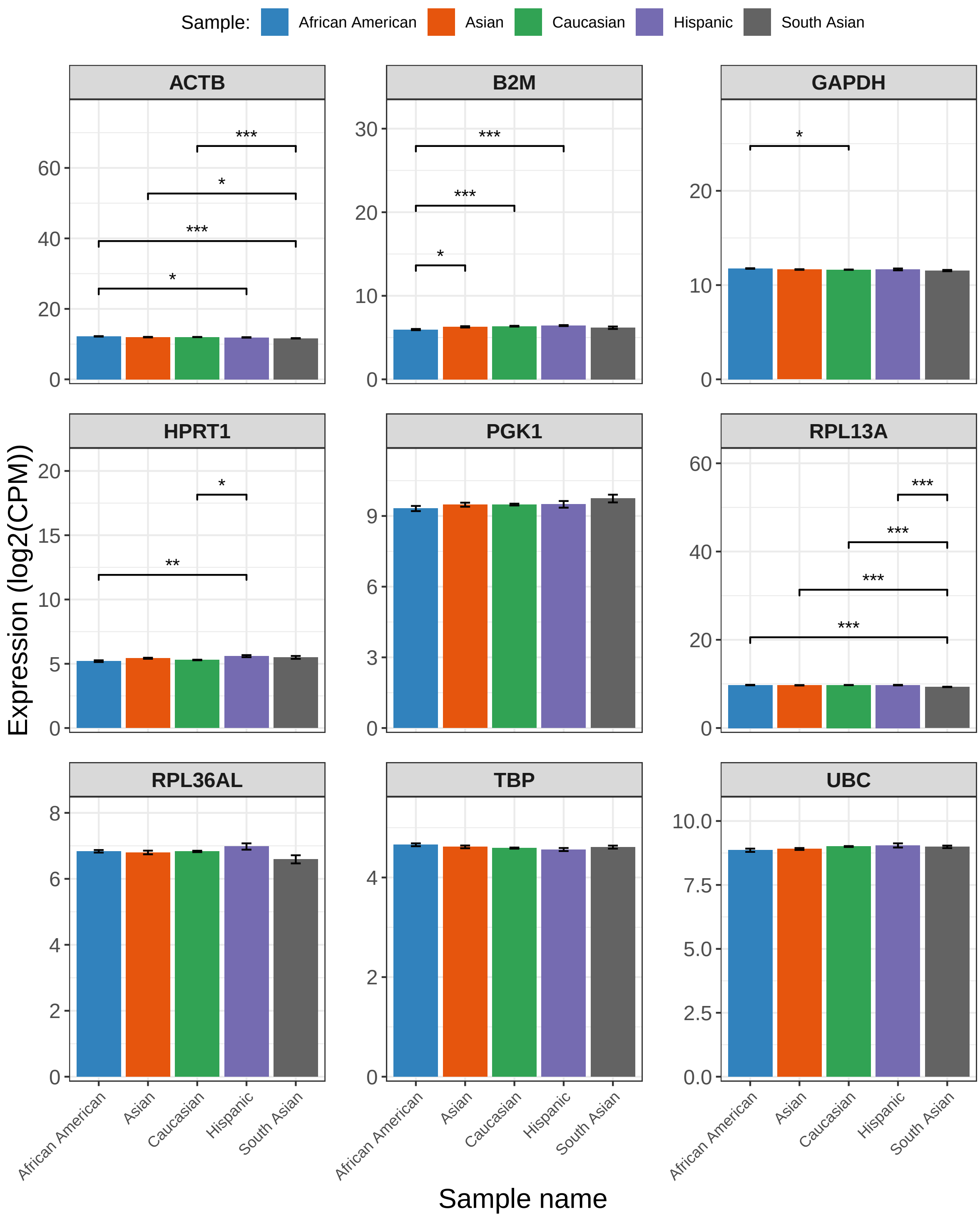


Figure 12: Expression of Housekeeping Genes by Race. Bar plots display log2 count per million gene expression by housekeeping gene. Bars represent mean expression ± standard error. Pairwise statistical significance was determined using Games-Howell post hoc tests following Welch's ANOVA, and significance was indicated with asterisks: $p < 0.05$ (∗), $p < 0.01$ (∗∗), $p < 0.001$ (∗∗∗).

the donor's race (Welch's ANOVA, $p < 0.001$). This suggests that environmental factors or ancestry linked biological factors are major contributors to transcriptional variability in iPSC lines.

To better understand the drivers of the observed PCA patterns, the top twenty genes with the greatest variability in expression across the iPSC samples were identified and labeled. These genes were primarily related to genes found on the sex chromosomes, such as male specific genes encoded on the Y chromosome: RPS4Y1 (var = 16.973), DDX3Y (var = 13.869), TXLNGY (var = 10.610), KDM5D (var = 10.160), USP9Y (var = 9.251), EIF1AY (var = 7.341), UTY (var = 7.231), NLGN4Y (var = 6.110); an X linked gene: RPS4XP22 (var = 5.800); and genes involved in X chromosome inactivation: XIST (var = 14.584) and TSIX (var = 12.091). Since the iPSC sample population was composed of a mix of female and male samples, this variability is to be expected, and was observed by Caramo-Orive *et al.* They found that genes with greater than 2% contributions to variation by sex were located on the X or Y chromosome (Carcamo-Orive et al., 2017).

The remaining nine genes with high variability in gene expression in iPSCs samples were associated with a variety of cell functions. CER1 (var = 7.325) and NLRP2 (var = 6.402) are associated with embryonic development and early differentiation, tumor suppressors MEG3 (var = 5.980) and LINC00261 (var = 5.554P), the transcription factors ZNF208 (car = 6.294), APOA1 (var = 6.096) involved in lipid transport, LARGE1 (var = 6.874), a gene involved in glycosylation in the Golgi apparatuss, CST1 (var = 9.794), a cysteine protease inhibitor , and COL3A1 (var = 6.400) for collagen type III, an extracellular matrix protein. Caramo-Orive *et al* investigated the top 200 genes with the most variability by performing a key driver analyses. They also found that LINC00261, CER1, and APOA2 (similar to APOA1) were key drivers of variability in iPSC samples, in addition to four other genes, GATA4, GATA6, EOMES, and FOXQ1 (Carcamo-Orive et al., 2017). These genes could be futher investigated to determine how their expression influences iPSC pluripotency and differentiation.

Pluripotency, Differentiation, and Houskeeping Gene Expression

The expression of pluripotency genes, SOX2, OCT4 (POU5F1), KLF4, c-Myc (MYC), TRA-1-81 and 1-60 (PODXL), NANOG, SSEA4 (FUT4), LIN28A, and Alkaline Phosphatase (ALPL)., were analyzed after grouping by both age-sex and race. These genes include transcription factors, SOX2 (SRY-box transciption factor 2), OCT4 (octamer-binding transcription factor 4), KLF4 (kruppel-like factor 4), and NANOG (nanog homeobox), involved in maintaining a pluripotent state. Additionally, the expression of PSC cell surface markers, SSEA4 (stage-specific embryonic antigen-4) and TRA-1-81/TRA-1-60 (tumor related antigen), an RNA binding protein, LIN28A (lin-28 homolog A), and a highly expressed enzyme in pluripotent cells, alkaline phosphatase, were analyzed. These genes have been found to be highly expressed in pluripotent cells, such as iPSCs (Takahashi et al., 2007).

To determine if the expression of these pluripotent genes varied across conditions, a linear mixed model was used. The predicted expression was modeled with fixed effects age decade, sex and race, with random intercepts grouped by gene to account for baseline differences in gene expression. In the equation, the observation of each gene is represented as 'o' , each gene is represented as 'g', and 'u' represents that the intercept is unique to each gene.

$$\hat{Expression}_{og} = \beta_0 + \beta_1(age\_decade_{og}) + \beta_2(sex_{og}) + \beta_3(race_{og}) + u_{0g} + \varepsilon$$

Fixed-effect significance was assessed using t-tests with Satterthwaite-approximated degrees of freedom to evaluate differences in gene expression relative to the reference groups (female, age 40s, African American).

Residual variance was 0.188 and gene intercept variance was 9.82, which suggests that the majority of the variance is between genes, not between groups.

It was found that there was no significant difference between female and male samples, but there was a 6.5% decrease in expression in 70s samples (β = - 0.097, $p < 0.05$). The 70s female group displayed a 6.5% decrease in pluripotency expression (β = - 0.097, $p < 0.05$.), whereas 70s male saw a 13% increase in pluripotency expression (β = + 0.179, $p < 0.001$). There was no observed difference between the other age-sex groups and the reference group (40s female), suggesting that iPSCs generated from donors in their 70s differ in their expression of pluripotency genes when compared to donors between the ages of 40 and 70. These results are observed in the Figure 4 top heatmap, which display higher expression in 70s males for several pluripotency genes (c-MYC, KLF4, SSEA4 (FUT4), and OCT4 (POU5F1)), and lower expression in several genes for 70s female (ALPL, LIN28A, SOX2, TRA-1-81/TRA-1-60 (PODXL), and c-MYC). When looking at race, iPSC samples from South Asian donors displayed a 10% decrease in pluripotency gene expression (β = -0.143, $p < 0.01$), but there was no difference between the other groups and the reference group, African Americans. This can be seen in Figure 4, where pluripotency expression for South Asian displayed both increased (SSEA4 (FUT4) and KLF4) and decreased expression (OCT4 (POU5F1), NANOG, LIN28A, SOX2, and ALPL) in the majority of pluripotency genes.

To look at the effect of individual genes, Welch's one-way ANOVA, assuming unequal variance, followed by Games-Howell post hoc tests with Benjamini-Hochberg corrected p values were used. SSEA4 (FUT4) displayed eleven significant pairwise comparisons across age-sex groups, LIN28A showed nine, and NANOG, POU5F1, and SOX2 had five significant pairwise comparisons, seen in Figure 5. KLF4 and ALPL showed the fewest significant differences, with only one pairwise comparison reaching significance, but all genes displayed significant differences in gene expression. For race, ALPL and LIN28A displayed five significant pairwise comparisons, and SOX2 displayed four, seen in Figure 6. Several pluripotency genes, including FUT4, NANOG, TRA-1-81 and 1-60 (PODXL), and OCT4 (POU5F1), did not show any significant pairwise differences across age-sex groups, indicating relatively stable expression.

Overall, pluripotency gene expresseion showed a higher variation in age-sex groups compared to race groups, suggesting that donor age has a greater effect on pluripotency gene expression in iPSC samples than sex and race.

Differentiation Markers

The expression of nine genes, which are commonly used as markers of differentiation, were analyzed. Three genes were selected for each of the three germ layers. For ectoderm markers, PAX6 (paired box 6) and Nestin (NES, neuroepithelial stem cell protein) were used as neural progenitor/stem cell markers, and KRT14 (keratin 14) as an epithelial marker. SOX17 (SRY-box transcription factor 17) was used as a general endoderm marker, AFP (alpha-fetoprotein) for hepatic cells (immature hepatocytes), and PDX1 (pancreatic and duodenal homeobox 1) as a pancreatic marker. For mesoderm markers, CD34 (cluster of differentation 34) for hematopoitic cells, CD90 (THY1, THYmocyte differentiation antigen 1) for mesenchymal cells, and NKX2-5 (NK2 homeobox 5) for cardiac progenitors were used. The expression of these linage specific markers is typically observed when populations of iPSCs have begun to randomly differentiation, and can be used to indicate the differentiation pathway of the cells.

The same model that was used with the pluripotency genes was used to analyze how age, sex, and race influence the expression of differentiation genes but substituting the pluripotency genes with differentiation genes.

Residual variance was 1.096 and gene intercept variance was 13.168, which again suggests that the majority of the variance is between genes, not between groups, however, these variances are larger than the variances observed with the pluripotency genes.

It was found that there was no difference in the expression of differentiation genes by sex, but there were statically significant differences across age and race groups. All three age groups, (50s, 60s, and 70s) displays differences in gene expression when compared to the reference group, 40s. iPSCs from donors in their 50s displays a 70.8% decrease (β = -1.748, $p < 0.001$), iPSCs from donors in their 60s displays a 70.6% decrease (β = -1.743, $p < 0.001$), and iPSCs from donors in thier 70s displayed a 57.7% increase (β = + 0.655, $p < 0.001$) in the expression of differentiation genes. This was further broken down by age-sex groups, in which 50s female had a 17% decrease (β = -0.269, $p < 0.001$), 50s male displayed a 24.5% decrease (β = -0.398, $p< 0.001$), and 60s female showed a 29.3% decrease, (β = -0.500, $p < 0.001$).

When looking at race, there was a 25.3% decrease in the expression of differentiation genes in iPSC samples from Asian donors (β = -0.429, $p < 0.001$), a 32.0% increase for Hispanic donors (β = + 0.399, $p < 0.01$), and an 80.0% increase for South Asian donors (β = + 0.854, $p < 0.001$). These differences can be seen by gene in the heatmap in Figure 7, in which the genes NKX2-5, PDX1, AFP, CD24, PAX6, and SOX17 showed high expression in South Asian samples, but low expression in Asian samples.

To look at the effect of individual genes, Welch's one-way ANOVA, assuming unequal variance, followed by Games-Howell post hoc tests with Benjamini-Hochberg corrected p values were used. KRT14, NKX2-5, and CD90 (THY1) displayed no statistically significant pairwise comparisons for both age-sex and race groups, seen in Figure 8 and 9, suggesting that there is minimal variation in the expression of these genes across the iPSC samples. The remaining genes displayed one to two significant pairwise comparisons (with p values < 0.05 and 0.01) for the age-sex groups, whereas these same genes displayed one to eight pairwise comparisons (with p values < 0.05, < 0.01, and 0.001) when samples were grouped by race. SOX17 and CD34, in particular displayed the most pairwise comparisons, at eight and six respectively. This suggests that the expression of differentiation markers is more heavily influenced by donor race rather than by age and sex.

Housekeeping Genes

The expression of nine commonly used reference/housekeeping genes in mammalian cultures, such as stem cells, were analyzed. GAPDH (glyceraldehyde-3-phosphate dehydrogenase) is one of the most widely used housekeeping genes since it is involved in glycolysis and is consistently expressed across various cell types (Panina et al., 2020).

Other commonly used gene involved in metabolism include PGK1 (phosphoglycerate kinase 1), cytoskelaton genes, ACTB (actin beta), and ribosomal genes, RPL36AL (ribosomal protein L36a Like), and RPL13A (ribosomal protein L13a). Other commonly used housekeeping genes include B2M (beta-2 microglobulin), which in volved in the formation of major histocompatibility complex (MHC) molecules, HPRT1 (hypoxanthine-guanine phosphoribosyltransferase 1) which is an enzyme involved in the purine salvage pathway, TBP for the TATA-box binding protein, and UBC for ubiquitin C.

Residual variance was 0.148 and gene intercept variance was 7.16, which suggests that the majority of the variance is between genes, not between groups. Additionally, this is the lowest variance observed between the three gene groups.

There was no difference in housekeeping expression of age groups 50s, 60s, and 70s when compared to the reference groups 40s, and a minimal increase in expression for male ($p < 0.1$). When looking at age-sex groups, there were some differences in housekeeping gene expression, with 9.6% increase in males in their 40s ($\beta = +\ 0.1325$, $p < 0.001$), a 2.4% increase for females in their 50s ($\beta = +\ 0.03375$, $p < 0.01$), and a 5.7% increase for females in their 60s ($\beta = +\ 0.08191$, $p < 0.01$). This can be seen in the top heatmap in Figure 10, where 40s males displayed higher expression of genes RPL36AL, GAPDH, RPL13A, and PGK1 and lower expression of ACTB and UBC, when compared to the reference groups, 40s female, which displayed lower expression in RPL36Al, GAPDH, RPL13A, and B2M and higher expression in ACTB and UBC.

When looking at race, iPSCs from Hispanic donors showed a 7.1% increase in housekeeping gene expression ($\beta = 0.09717$, $p < 0.05$), but there was no statistically significant differences between the other groups when compared to the reference group, African American. When looking at the bottom heatmap in Figure 10, Asian and Caucasian samples displayed consistent average expression values around zero, suggesting that there is minimal variation in expression between housekeeping genes for these samples. African American and South Asian samples diaplayed the more variation, with a decrease in the expression for PGK1, HPRT1, B2M, and UBC and increased expression of TBP, ACTB, and GAPDH for African American samples, and decreased expression of ACTB, GAPDH, RPL13A, and RPL36AL and increased expression of PDK1 for South Asian samples.

To look at the effect of individual genes, Welch's one-way ANOVA, assuming unequal variance, followed by Games-Howell post hoc tests with Benjamini-Hochberg corrected p values were used. ACTB and RPl36AL displayed no pairwise comparisons when grouping by age-sex, B2M, GAPDH, HPRT1, TBP, RPL13A, and UBC displayed two to five significant comparisons, and PGK1 displayed ten significant comparisons. This suggests that PGK1 expression varies significantly when grouping by age-sex, however, there were no significant comparisons when grouping by race. RPL36AL, TBP, and UBC also displayed no significant comparisons when grouping by race. ACTB, B2M, GAPDH, HPRT1, and RPL13A displayed one to four comparisons.

Across all nine genes, RPL36AL is the only gene that displayed no statistically significant pairwise comparisons for both age-sex groups and race, suggesting that the expression of this gene is not influenced by age, sex, or race and is consistently expressed across all of the anazlyed iPSC samples. Overall, it appears that age and sex are the main drivers of variation in housekeeping gene expression of the groups studied here.

This is supported by the study conducted by Panina *et al* who investigated the expression of 70 commonly used housekeeping genes during iPSC reprogramming. They found that some of the most commonly used genes, GAPDH, ACTB, HPRT, and B2M, fluctuated more than three fold (Panina et al., 2020). They also round that ribsosomal genes, such as RPL13A, were the most stable regarding their expression during the reprogramming process. In this study RPL13A expression appears to fluctuate in iPSC samples, however, RPL36AL does not appear to.

Conclusion

The transcriptome of 317 iPSC samples, collected using RNA-Seq from a variety of donors, were analyzed to characterize overall RNA composition, identify key drivers of expression variability via principal component analysis (PCA), and access variability in the expression of specific pluripotency, differentiation, and reference genes. While a variety of RNA types were detected, 91.7% of the total expression was contributed by protein coding mRNAs. PCA and Welch's one-way ANOVA revealed that donor age-sex groups and race were major contributors to expression variability, with the top genes driving sample seperation including sex-linked genes located on the X and Y chromosomes.

Expression of commonly used pluripotency markers, including LIN28A and SOX2, varied across age–sex and race groups. Similarly, differentiation markers such as SOX17 and CD34 exhibited numerous significant pairwise differences, particularly between racial groups. Even housekeeping genes, which are typically assumed to be stably expressed, displayed variability across age–sex and race groups, notably GAPDH, HPRT1, and PGK1. These findings indicate that donor demographic characteristics can influence the transcriptome of iPSCs, including genes often used as references or markers. However, because group sizes were unequal, some variability may reflect limited sample numbers, even when using Welch's ANOVA with Benjamini-Hochberg p-value correction. Together, these results demonstrate that donor demographics, including age, sex, and race, significantly influence global gene expression patterns in iPSCs, affecting both highly variable and commonly used reference genes.

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